

Physics Framework for 0D Global Model (Argon Plasma)

Species Considered

- Electrons (e^-)
- Ar (ground state neutral)
- Ar* (metastable, 11.55 eV)
- Ar⁺ (singly charged ion)

Key Reactions

#	Reaction	Threshold	Rate Coefficient
1	$e + \text{Ar} \rightarrow e + e + \text{Ar}^+$	15.76 eV	$K_{iz}(\text{Te})$ — from cross section
2	$e + \text{Ar} \rightarrow e + \text{Ar}^*$	11.55 eV	$K_{ex}(\text{Te})$ — from cross section
3	$e + \text{Ar}^* \rightarrow e + e + \text{Ar}^+$	4.21 eV	$K_{iz}^*(\text{Te})$ — from cross section
4	$\text{Ar}^* + \text{Ar}^* \rightarrow \text{Ar}^+ + \text{Ar} + e$	—	$K_{\square\square} \approx 5 \times 10^{-16} \text{ m}^3/\text{s}$
5	$\text{Ar}^* \rightarrow \text{Ar}$ (wall quenching)	—	Loss via diffusion
6	$\text{Ar}^+ \rightarrow \text{wall}$ (Bohm flux)	—	$n_{\square uB} = n_{\circ hluB}$

Governing Equations

1. Electron Temperature — Particle Balance

$$K_{iz}(T_e) \cdot n_g \cdot d_{eff} = u_B(T_e)$$

where:

- $u_B = \sqrt{eT_e/M}$ — Bohm velocity
- $d_{eff} = \frac{Rl}{2(Rh_l + lh_R)}$ — effective plasma size
- h_l, h_R — edge-to-center density ratios (Lieberman & Lichtenberg form)

$$h_l = 0.86 \left(3 + \frac{l}{2\lambda_i} \right)^{-1/2}, \quad h_R = 0.80 \left(4 + \frac{R}{\lambda_i} \right)^{-1/2}$$

2. Plasma Density — Energy Balance

$$n_0 = \frac{P_{abs}}{e \cdot u_B \cdot A_{eff} \cdot E_T}$$

where:

- $A_{eff} = 2\pi R^2 h_l + 2\pi Rl h_R$
- $E_T = E_c + E_i + E_e$ — total energy lost per ion-electron pair lost
- E_c — collisional energy loss (from cross sections)
- $E_i = eV_s + \frac{1}{2}T_e$ — mean ion bombarding energy
- $E_e = 2T_e$ — mean electron energy lost



3. Collisional Energy Loss per Ion-Electron Pair

$$E_c = \frac{\sum_j \epsilon_j K_j(T_e)}{\sum_j K_j^{ion}(T_e)}$$

summing over all inelastic collisions weighted by their energy threshold ϵ_j .

4. Metastable Density Balance

$$\frac{dn_{Ar^*}}{dt} = K_{ex}n_en_g - K_{iz}^*n_en_{Ar^*} - K_{ss}n_{Ar^*}^2 - \frac{n_{Ar^*}}{\tau_*} = 0$$

Solved algebraically in a steady state.

5. Rate Coefficients from Cross Sections (EEDF Integration) - to extract cross section data from Lxcat file

For Maxwellian EEDF:

$$K_j(T_e) = \sqrt{\frac{2}{m_e}} \cdot \left(\frac{1}{k_B T_e} \right)^{3/2} \cdot \frac{4\pi}{(4\pi)^{1/2}} \int_{\epsilon_{th}}^{\infty} \epsilon \cdot \sigma_j(\epsilon) \cdot f_M(\epsilon, T_e) d\epsilon$$

Using energy-space form:

$$K_j(T_e) = \sqrt{\frac{2e}{m_e}} \int_0^{\infty} \sigma_j(\epsilon) \cdot \epsilon \cdot \frac{2}{\sqrt{\pi}(k_B T_e)^{3/2}} e^{-\epsilon/k_B T_e} d\epsilon$$

Output Parameters

Parameter	Symbol
Electron temperature	T _e (eV)
Plasma density	n ₀ (m ⁻³)
Ion current density	J _i (mA/cm ²)
Ion bombarding energy	E _i (eV)
Collisional energy loss	E _c (eV)
Bohm velocity	u _B (m/s)
Metastable density	n* (m ⁻³)

How to use LXCat file

The parser automatically detects cross-section blocks by looking for keywords like **IONIZATION**, **EXCITATION**, **ELASTIC** in the headers. Standard LXCat **.txt** exports (from www.lxcat.net) work directly, no reformatting needed. If a process is missing in file, it falls back to an analytic approximation.

What the code computes

Step-by-step physics:

1. Converts pressure $\rightarrow$ neutral density **ng**, then computes **λ_i** , **h1**, **hR**, **deff**
2. **Particle balance** $\rightarrow$ solves **$K_{iz}(T_e) \cdot ng \cdot deff = uB(T_e)$** numerically for **Te**
3. **Rate coefficients** $\rightarrow$ integrates **$K(T_e) = \int \sigma(\epsilon) \cdot v(\epsilon) \cdot fM(\epsilon, T_e) d\epsilon$** over your LXCat cross sections using a Maxwellian EEDF
4. **Energy balance** $\rightarrow$ **$n_0 = P_{abs} / (e \cdot uB \cdot A_{eff} \cdot ET)$**
5. **Metastable density** $\rightarrow$ solves the quadratic steady-state balance for **n***
6. Outputs **Te**, **n0**, **n***, **Ji**, **Ei**, **Ec**, **ET**

What we get

- Console printout replicating **textbook Example 1** at $P = 800$ W, 1 mTorr
- **6-panel figure**: each parameter vs absorbed power, with a **Te** vs pressure inset
- **2D heatmaps**: **n0**, **Te**, **Ji** as colour maps over the full (P , p) space

What's in file

Process	Count	Details
Elastic ($e + \text{Ar} \rightarrow e + \text{Ar}$)	1	Biagi/Magboltz momentum transfer, complete set
Excitation ($e + \text{Ar} \rightarrow e + \text{Ar}^*$)	44	Full resolved spectrum, 11.55 eV to 15.66 eV
Ionization ($e + \text{Ar} \rightarrow 2e + \text{Ar}^+$)	1	Threshold 15.76 eV
Ar⁺ + Ar charge-exchange	1	Phelps isotropic, from Ar ⁺ perspective

From the paper - Global model of Ar, O₂, Cl₂, and Ar/O₂ high-density plasma discharges C. Lee; M. A. Lieberman

Background: 0D Global Model for Argon Discharges

What is a Global (0D) Model?

A global model, also called a volume-averaged or 0D model, treats the entire plasma reactor as a single well-mixed control volume. Rather than solving spatially resolved PDEs (as in fluid or PIC simulations), it solves a coupled set of **algebraic/ODE equations** for:

- **Electron temperature** T_e (from power balance)
- **Plasma density** n_i (from particle balance)
- **Metastable density** n^* (from particle balance)

This makes it computationally cheap while capturing the essential physics of the discharge.

Physics of the Argon Global Model

1. Reaction Set (from Table III of the paper)

Reaction	Description	Rate Coefficient
$e + \text{Ar} \rightarrow \text{Ar}^+ + 2e$	Direct ionization	$k_1 = 1.23 \times 10^{-7} \exp(-18.68/T_e)$
$e + \text{Ar} \rightarrow \text{Ar}^* + e$	Excitation to metastable	$k_2 = 3.71 \times 10^{-8} \exp(-15.06/T_e)$
$e + \text{Ar}^* \rightarrow \text{Ar}^+ + 2e$	Step ionization	$k_3 = 2.05 \times 10^{-7} \exp(-4.95/T_e)$
$e + \text{Ar}^* \rightarrow \text{Ar} + e$	Superelastic quenching	$k_4 = 2.0 \times 10^{-7}$
$\text{Ar}^* + \text{Ar}^* \rightarrow \text{Ar} + \text{Ar}^+$	Metastable pooling	$k_5 = 6.2 \times 10^{-10}$

2. Power Balance

The absorbed power must equal all energy loss channels:

$$P_{\text{abs}} = P_{\text{ev}} + P_{\text{iw}} + P_{\text{ew}}$$

Where:

- P_{ev} = electron energy loss in volume = $n_e \cdot V \cdot \sum n_i \cdot k_i \cdot \epsilon_i$
- P_{iw} = ion energy to walls = $n_i \cdot u_B \cdot A \cdot \epsilon_{\text{iw}}$, with $\epsilon_{\text{iw}} \approx (5-8)T_e$
- P_{ew} = electron energy to walls = $n_e \cdot u_B \cdot A \cdot 2T_e$

The collisional energy loss per electron-ion pair ϵ_L (eV) encodes all inelastic channels.

3. Particle Balance for Ar^+

$$Vn_e(k_1 n_{Ar} + k_3 n_{Ar^*}) = A_{eff} \cdot n_{is} \cdot u_B$$

Ions created in volume = ions lost to walls at Bohm velocity $u_B = \sqrt{(eT_e/M)}$.

The sheath-edge to bulk density ratio (from Appendix of paper):

$$hL = 0.863 + L/2\lambda$$

4. Particle Balance for Ar* (Metastables)

$$Vn_e k_2 n_{Ar} = Vn_e(k_3 + k_4)n_{Ar^*} + V k_5 n_{Ar^*}^2 + L^2 D_{eff} n_{Ar^*} V$$

5. Self-Consistent Solution

Given P_{abs} , pressure p , and geometry (L , R):

1. Guess T_e
2. Compute all rate coefficients $k_i(T_e)$
3. Solve metastable balance for n^*
4. Check particle balance — adjust T_e until satisfied
5. Use power balance to find n_e

Physics approach:

- Parse each cross section block (ELASTIC, EXCITATION, IONIZATION) from the LXCat file
- For each process, compute the rate coefficient by integrating $\sigma(\epsilon)$ over a Maxwellian EEDF:
- Map the resulting $k(T_e)$ functions to the roles previously filled by k_1 – k_5 in the global model
- Everything else (particle balance, power balance) stays the same

Finally from the paper - On the effect of the electron energy distribution on the plasma parameters of an argon discharge: a global (volume-averaged) model study , J T Gudmundsson 2001 Plasma Sources Sci. Technol. 10 76

Section	What it explains
Background	Why EEDF matters; why real discharges deviate from Maxwellian; role of Druyvesteyn

Global model	The two balance equations (particle + power) and what each solves for
Generalised EEDF	The $f(E) = c_1 E^{1/2} \exp(-c_2 E^x)$ family, how c_1/c_2 are derived from Γ functions
Ion velocity & sheath	Generalised Bohm criterion, floating potential V_s , energy terms E_e and E_i
Sheath edge density ratios	h_L and h_R formulae, physical meaning, σ_{in} value and its source
Collisional energy loss	E_c breakdown: ionisation + all 5 excitation groups + elastic losses
Cross sections	Which data source is used for each process and how it's implemented
Reactor geometry	All numerical values with units
Key results	Physical interpretation of each reproduced figure
References	Numbered exactly as in the paper